ENGINEERING REFERENCE / REVIEWED EDITION

Mechanics of Materials

I & II

Formula Book

Strength of materials, from axial loading to finite elements.

26 topic sheets with equations, symbols, model limits and worked checks.

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Companion to cad-automation-inventor-plant3d

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How to use the book

Read the symbols and the model assumptions before substituting numbers. The boxed worked check lets you confirm units and magnitudes. This is a reference for the listed two-course syllabus, not a complete structural design standard.

Consistent units

Quantity	SI	Convenient part calculations
Force	N	N
Length	m	mm
Stress / modulus	Pa	MPa = N/mm²
Moment / torque	N m	N mm
Area moment / torsion constant	m⁴	mm⁴
Energy	J	N mm; divide by 1000 for J
Strain	dimensionless	microstrain = 10^{-6}
Rotation	radians	radians

Sign and model conventions

- Normal stress is tensile-positive; compression is negative, except where a compressive load magnitude is explicitly used.
- Beam coordinate x increases right; load w is downward-positive; M is sagging-positive; v is upward-positive.
- Tables give positive magnitudes where stated. Do not transfer their sign into a different convention without checking.
- Engineering shear strain is twice tensor shear strain.
- J for circular torsion equals the polar area moment; Jt for a noncircular section generally does not.
- Use all three principal stresses in 3D failure checks, including zero for plane stress.

A reliable calculation

Sketch the geometry and supports. Find internal loads. Select the governing model. Keep units consistent. Calculate stress and deformation. Check equilibrium and a limiting case. For real design, apply the relevant material data, load combinations and standards.

Scope and revision

26 topics cover Mechanics of Materials I and II. Version: 13 November 2026, reviewed formula-book edition. Formula source, detailed chapters, books and correction notes are available in the repository.

Equilibrium and distributed loads

Symbols on this sheet

F: external force; M: moment about a stated point; r: lever-arm vector; w: force per length; W: resultant force; x: position. Resultant location is undefined if W = 0.

$$\sum \mathbf{F} = 0 \qquad \sum \mathbf{M} = 0$$

$$\mathbf{M} = \mathbf{r} \times \mathbf{F} \qquad W = \int w\,dx$$

$$\bar{x} = \frac{\int x w\,dx}{W}$$

A uniform load has $W = wL$; a triangular load has $W = w_{\max}L/2$ and acts at $L/3$ from its larger end.

Worked check

A simply supported beam has span 4 m and a downward 6 kN load 1 m from its left support. Taking moments about the left support gives $R_B(4) = 6(1)$, so $R_B = 1.5$ kN. Vertical equilibrium gives $R_A = 4.5$ kN. Check that both reactions total 6 kN.

When to use this model

Draw a cut before using any stress equation. An incorrect internal load cannot be fixed by a more advanced stress formula. Statics assumes negligible acceleration; dynamics needs additional equations.

Open the full chapter and practice question

Average stress and strain

Symbols on this sheet

N: axial force; V: shear force; A: section area; As: shear area; m: number of equally loaded shear planes; d: pin/hole diameter; t: plate thickness; delta: extension; L: gauge length.

$$\sigma = \frac{N}{A} \qquad \tau_{\text{avg}} = \frac{V}{A_s}$$

$$\tau_{\text{pin}} = \frac{P}{m\pi d^2/4} \qquad \sigma_{\text{bearing}} = \frac{P}{td}$$

$$\epsilon = \frac{\delta}{L} \qquad \gamma \approx \frac{\Delta u}{h}$$

$$A_{\text{net}} = (b - d)t$$

Average stress is not the local peak at a hole or contact.

Worked check

A 20 mm diameter rod carries 30 kN tension. Its area is $\pi 20^2/4 = 314.159$ mm², giving $\sigma = 95.493$ MPa. If a 500 mm gauge length extends 0.25 mm, $\epsilon = 0.0005 = 500$ microstrain.

When to use this model

Actual connection stresses are nonuniform. Bearing, net-section tension, shear-out, pin bending and joint geometry can all matter. The average equations are first models, not complete joint design rules.

Open the full chapter and practice question

Material laws

Symbols on this sheet

E: Young's modulus; G: shear modulus; K: bulk modulus; nu: Poisson ratio; Sy: yield strength; A0 and L0: original dimensions. Energy per volume has stress units.

$$\sigma = E\epsilon \qquad \tau = G\gamma$$

$$G = \frac{E}{2(1+\nu)} \qquad K = \frac{E}{3(1-2\nu)}$$

$$\nu = -\frac{\epsilon_{\text{lateral}}}{\epsilon_{\text{axial}}}$$

$$w_e = \frac{\sigma^2}{2E} \qquad w_r = \frac{S_y^2}{2E}$$

$$\sigma_t = \frac{P}{A_{\text{current}}} \qquad \epsilon_t = \ln\frac{L}{L_0}$$

The scalar normal-stress law and energy-density formula above are uniaxial; multiaxial elasticity is in II-14. Under uniform constant-volume deformation, $\sigma_t = \sigma_e(1+\epsilon_e)$.

Engineering measures use the original area and length:

$$\sigma_e = \frac{P}{A_0} \qquad \epsilon_e = \frac{L - L_0}{L_0}$$

The elastic Poisson range is $-1 < \nu < 0.5$. The shear and bulk modulus conversions require an isotropic material.

Worked check

With $E = 200$ GPa and $\nu = 0.30$, $G = 76.923$ GPa and $K = 166.667$ GPa. At uniaxial stress 100 MPa, axial strain is 0.0005 and lateral strain is −0.00015.

When to use this model

A nominal tensile yield factor is $n = S_y/|\sigma|$ for a uniaxial check. Material grade, temperature, processing and loading rate affect properties; example values here are teaching inputs.

Open the full chapter and practice question

Axial and thermal deformation

Symbols on this sheet

N(x): signed axial force; E(x), A(x): local modulus and area; alpha: thermal expansion coefficient; delta T: temperature change; rho: mass density; g: gravity; k: axial stiffness.

$$\delta = \int \frac{N}{EA}\,dx \qquad \delta = \sum_i \frac{N_i L_i}{E_i A_i}$$

$$k_{\text{axial}} = \frac{EA}{L}$$

$$\epsilon_T = \alpha \Delta T \qquad \delta_T = \alpha L \Delta T$$

$$\sigma_{\text{fully restrained}} = -E\alpha\Delta T$$

For a uniform hanging bar under self-weight:

$$\sigma_{\text{top}} = \rho g L \qquad \delta = \frac{\rho g L^2}{2E}$$

Worked check

For $N = 20$ kN, $L = 1000$ mm, $A = 200$ mm² and $E = 200000$ MPa, mechanical extension is 0.5 mm. If $\alpha = 12 \times 10^{-6}$/K and $\Delta T = 40$ K, free thermal extension is 0.48 mm. With the same applied load and free thermal movement, total extension is 0.98 mm.

When to use this model

For varying temperature use $\int \alpha(x)\Delta T(x)\,dx$. Gaps and yielding change the active constraints; determine whether contact occurs before imposing full restraint.

Open the full chapter and practice question

Compatibility

Symbols on this sheet

ki: member axial stiffness Ei Ai/Li; Pi: member force; delta: common displacement. Parallel-bar load sharing assumes rigid end plates, engaged members and compatible axial movement.

Parallel engaged bars: $\delta = P/\sum k_i$, $P_i = k_i\delta$. Series compliance: $1/k_{\text{eq}} = \sum 1/k_i$. For two bonded equal-length thermally loaded bars:

$$A_1\sigma_1 + A_2\sigma_2 = P \qquad \frac{\sigma_1}{E_1} + \alpha_1\Delta T = \frac{\sigma_2}{E_2} + \alpha_2\Delta T$$

Worked check

Two equal-length bars support 60 kN through rigid end plates. Bar 1 has $E_1 = 200000$ MPa, $A_1 = 200$ mm²; bar 2 has $E_2 = 70000$ MPa, $A_2 = 400$ mm². Their stiffness ratio is $40 : 28$. Thus $P_1 = 35.294$ kN and $P_2 = 24.706$ kN. Their stresses are 176.471 and 61.765 MPa.

When to use this model

Write geometry and compatibility before substituting numbers. Equal force, equal stress and equal strain are different statements. Initial gaps, different lengths, slip, temperature and plasticity can invalidate simple load-sharing fractions.

Open the full chapter and practice question

Section geometry

Symbols on this sheet

Ix, Iy: second moments about named centroidal axes; Ixy: product moment; d: parallel-axis offset (diameter in the shape table); c: extreme-fiber distance; rg: radius of gyration.

$$\bar{y} = \frac{\sum A_i y_i}{\sum A_i} \qquad I_x = \int y^2 dA$$

$$I_y = \int x^2 dA \qquad I_{xy} = \int xy\, dA$$

$$I = I_c + Ad^2 \qquad J_p = I_x + I_y$$

$$Z = \frac{I}{c} \qquad r_g = \sqrt{\frac{I}{A}}$$

Section	Area	Centroidal I	Circular torsion J
Rectangle $b \times h$	bh	$bh^3/12$ about horizontal axis	Do not use $I_x + I_y$
Circle diameter d	$\pi d^2/4$	$\pi d^4/64$	$\pi d^4/32$
Annulus D, d	$\pi(D^2 - d^2)/4$	$\pi(D^4 - d^4)/64$	$\pi(D^4 - d^4)/32$

Triangle of base b, height h: $A = bh/2$, centroid $h/3$ from base, centroidal $I = bh^3/36$ about an axis parallel to the base.

Worked check

A rectangle has $b = 40$ mm and $h = 80$ mm. $I_x = 1.706667 \times 10^6$ mm⁴ and $Z_x = 42666.7$ mm³. Rotating it through 90° gives $I_x = 426666.7$ mm⁴, one quarter as large.

When to use this model

Always sketch the actual bending axis. A catalog's strong-axis property cannot be used for loading about the weak axis. For composite shapes find the centroid first, then apply the parallel-axis theorem.

Open the full chapter and practice question

Circular torsion and power

Symbols on this sheet

T: torque; r: radial location; J: polar moment for a circular section; G: shear modulus; phi: twist in radians; n: rpm; omega: rad/s; power is in W when T is N m.

$$\tau = \frac{Tr}{J}$$

$$\tau_{\max,\text{solid}} = \frac{16T}{\pi d^3}$$

$$\phi = \int \frac{T}{GJ}\,dx$$

$$\mathcal{P} = T\omega$$

$$\omega = \frac{2\pi n}{60}$$

Angles are radians; n is rpm.

Worked check

A solid shaft has $d = 40$ mm, $L = 1000$ mm and $G = 80000$ MPa. At $T = 500$ N·m, $J = 251327.4$ mm⁴, $\tau_{\max} = 39.789$ MPa and $\phi = 0.024868$ rad = 1.425°. At 1200 rpm, the transmitted power is 62.832 kW.

When to use this model

For fixed-ended shafts, combine torque equilibrium with twist compatibility. These circular-shaft stress equations do not apply to rectangles or open channels. Keyways and shoulders introduce local stress concentrations; bearing reactions and gear loads can add bending.

Open the full chapter and practice question

Beam internal loads

Symbols on this sheet

V: internal shear; M: internal bending moment; w: downward load per length; x: coordinate to the right; L: span; a and b: distances from point load to the two supports.

$$V' = -w \qquad M' = V$$

for downward-positive w and sagging-positive M.

Load case	Maximum moment magnitude
Cantilever, tip force P	PL at fixed end
Cantilever, full-span UDL w	$wL^2/2$ at fixed end
Simply supported, central P	$PL/4$
Simply supported, point P at distances a, b from supports, $a + b = L$	Pab/L
Simply supported, full-span UDL w	$wL^2/8$

For the off-center point load, left reaction Pb/L and right reaction Pa/L.

Worked check

For a simply supported span $L = 4$ m carrying $w = 2$ kN/m throughout, each reaction is 4 kN.

$$V(x) = 4 - 2x \text{ kN} \qquad M(x) = 4x - x^2 \text{ kN m}$$

At $x = 2$ m, $V = 0$ and $M_{\max} = 4$ kN·m. At both supports the moment is zero. The area under V from 0 to 2 m is 4 kN·m, confirming the moment increase.

When to use this model

Split piecewise problems at every load, support and load-distribution change. Sketch both diagrams with units. A fixed support usually has a nonzero moment; “moment is zero at a support” applies to the appropriate ideal pin/roller condition, not all supports.

Open the full chapter and practice question

Elastic bending

Symbols on this sheet

M: bending moment; y: coordinate from the neutral axis, positive upward; I: centroidal principal second moment; Z: elastic section modulus; N: signed axial force.

$$\sigma_x = -\frac{My}{I} \qquad |\sigma_{\max}| = \frac{|M|}{Z}$$

$$Z_{\text{required}} = \frac{|M|}{\sigma_{\text{allow}}} \qquad \sigma_{\text{axial+bending}} = \frac{N}{A} - \frac{My}{I}$$

Use centroidal principal axes for the simple form.

Worked check

A rectangular beam is 40 mm wide and 80 mm deep with $M = 2$ kN·m. Chapter 6 gives $I = 1.706667 \times 10^6$ mm⁴ and $c = 40$ mm. The extreme stress magnitude is 46.875 MPa. One face is in tension and the other in compression.

When to use this model

The ordinary neutral axis passes through the centroid only under the stated homogeneous elastic assumptions. Different bonded materials require an elastic-modulus-weighted section analysis. Deep beams, curved beams, local load application, yielding and lateral-torsional buckling require additional models. Do not treat a peak near a sharp notch as the nominal Mc/I value.

Open the full chapter and practice question

Beam shear and shear flow

Symbols on this sheet

Q: first moment of the selected partial area about the neutral axis; b: local width; q: shear flow in force/length; s: connector spacing. Q is not the second moment I.

$$Q = \int_{A'} y\,dA \qquad \tau = \frac{VQ}{Ib}$$

$$q = \frac{VQ}{I} \qquad F_{\text{station}} = qs$$

Rectangle: $\tau_{\max} = 3V/(2A)$. Solid circle: $\tau_{\max} = 4V/(3A)$.

Worked check

A 40 × 80 mm rectangular beam carries $V = 10$ kN. Its area is 3200 mm². Average shear is 3.125 MPa and maximum shear is 4.6875 MPa.

When to use this model

For an I-section, calculate Q and local web thickness at the web level. A flange-web junction transfers longitudinal shear even though it may look “horizontal.” Thin-walled unsymmetric sections can twist under transverse load; chapter 19 introduces the shear center.

Open the full chapter and practice question

Elastic deflection

Symbols on this sheet

v: upward-positive deflection; theta: slope; E: modulus; I: bending-axis second moment; L: beam length. P and w are downward load magnitudes in the tables.

For upward-positive v: $EIv'' = M$ and $(EIv'')'' = -w$.

Case, constant EI	Maximum deflection magnitude	Slope magnitude
Cantilever tip P	$PL^3/(3EI)$	$PL^2/(2EI)$ at tip
Cantilever full UDL w	$wL^4/(8EI)$	$wL^3/(6EI)$ at tip
Simply supported central P	$PL^3/(48EI)$	$PL^2/(16EI)$ at support
Simply supported full UDL w	$5wL^4/(384EI)$	$wL^3/(24EI)$ at support

Moment-area relations with compatible signed curvature:

$$\theta_B - \theta_A = \int_A^B \frac{M}{EI}\,dx$$

$$v_B - v_A - \theta_A(B - A) = \int_A^B (B - x)\frac{M}{EI}\,dx$$

Worked check

A cantilever has $P = 1000$ N, $L = 1000$ mm, $E = 200000$ MPa and $I = 2 \times 10^6$ mm⁴. Tip deflection magnitude is 0.833333 mm and tip slope is 0.00125 rad.

When to use this model

Superpose signed responses only when linearity and unchanged support conditions apply. The moment-area relations integrate M/EI to obtain slope change and tangential deviation. Shear deformation matters for short/deep beams; see chapter 16.

Open the full chapter and practice question

Thin vessels and springs

Symbols on this sheet

p: internal-minus-external pressure; r: vessel mean radius; t: wall thickness. For springs D: mean coil diameter; d: wire diameter; na: active turns; Cs: spring index.

$$\sigma_h = \frac{pr}{t} \qquad \sigma_{z,\text{closed}} = \frac{pr}{2t}$$

$$\sigma_{\text{sphere}} = \frac{pr}{2t}$$

$$\epsilon_h = \frac{\sigma_h - \nu\sigma_z}{E} \qquad \epsilon_z = \frac{\sigma_z - \nu\sigma_h}{E}$$

$$\frac{\Delta V}{V} \approx 2\epsilon_h + \epsilon_z$$

The volume relation is for the small strain of a cylindrical volume. Helical spring:

$$\delta = \frac{8PD^3 n_a}{Gd^4} \qquad k = \frac{Gd^4}{8D^3 n_a}$$

$$C_s = \frac{D}{d} \qquad K_w = \frac{4C_s - 1}{4C_s - 4} + \frac{0.615}{C_s}$$

$$\tau_{\max} \approx K_w \frac{8PD}{\pi d^3}$$

For a free thin sphere in an elastic membrane state, tangential strain is $\epsilon_t = (1-\nu)\sigma_{\text{sphere}}/E$; to first order $\Delta r/r = \epsilon_t$ and $\Delta V/V = 3\epsilon_t$. The cylinder formulas presume pressure thrust carried by closed ends. Spring equations assume a close-coiled round-wire spring; n_a counts active turns and D is the mean coil diameter.

Worked check

Example: $p = 2$ MPa, $r = 100$ mm, $t = 5$ mm gives hoop stress 40 MPa and closed-end axial stress 20 MPa.

Example: $P = 100$ N, $D = 40$ mm, $d = 5$ mm, $n_a = 8$, $G = 80000$ MPa gives $\delta = 8.192$ mm and $k = 12.207$ N/mm.

When to use this model

Use the assumptions stated beside the formulas and the linked chapter.

Open the full chapter and practice question

Stress transformation

Symbols on this sheet

sigma x, sigma y, tau xy: stress components at one point; theta: counterclockwise axes rotation; C: circle center; R: circle radius. Use atan2 to retain quadrant information.

$$C = \frac{\sigma_x + \sigma_y}{2} \qquad R = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$\sigma_{x'} = C + \frac{\sigma_x - \sigma_y}{2}\cos 2\theta + \tau_{xy}\sin 2\theta$$
$$\tau_{x'y'} = -\frac{\sigma_x - \sigma_y}{2}\sin 2\theta + \tau_{xy}\cos 2\theta$$

$$\sigma_{1,2} = C \pm R \qquad 2\theta_p = \operatorname{atan2}(2\tau_{xy}, \sigma_x - \sigma_y)$$
$$\tau_{\max,3D} = \frac{\sigma_{\max} - \sigma_{\min}}{2}$$

Include the third principal stress, zero for plane stress, in the last expression.

The in-plane maximum shear is $\tau_{\max,\text{in}} = R$; it need not equal the absolute 3D maximum. Re-sort the two in-plane principal stresses together with zero for plane stress.

Worked check

At $\sigma_x = 80$, $\sigma_y = 20$, $\tau_{xy} = 30$ MPa, $C = 50$ and $R = 42.426$ MPa. The in-plane principal stresses are 92.426 and 7.574 MPa. Absolute maximum shear is 46.213 MPa, not 42.426 MPa, because zero is the third principal stress.

When to use this model

Mohr's circle has center C and radius R; physical rotations map to twice the angle. Published plots differ in shear-axis direction. State the plotting convention and verify one transformed stress rather than guessing the rotation direction.

Open the full chapter and practice question

Strain and 3D elasticity

Symbols on this sheet

epsilon0, epsilon45, epsilon90: measured directional strains; gamma xy: engineering shear strain; E, nu, G: isotropic constants; alpha delta T: stress-free thermal strain.

$$\epsilon_\theta = \frac{\epsilon_x + \epsilon_y}{2} + \frac{\epsilon_x - \epsilon_y}{2}\cos 2\theta + \frac{\gamma_{xy}}{2}\sin 2\theta$$

$$\epsilon_x = \epsilon_0 \qquad \epsilon_y = \epsilon_{90}$$
$$\gamma_{xy} = 2\epsilon_{45} - \epsilon_0 - \epsilon_{90}$$

$$\epsilon_x = \frac{\sigma_x - \nu(\sigma_y + \sigma_z)}{E} + \alpha\Delta T \qquad \gamma_{xy} = \frac{\tau_{xy}}{G}$$

Use cyclic versions for other directions. For plane stress without thermal strain:

$$\sigma_x = \frac{E}{1-\nu^2}(\epsilon_x + \nu\epsilon_y) \qquad \sigma_y = \frac{E}{1-\nu^2}(\epsilon_y + \nu\epsilon_x)$$

The in-plane principal strains are

$$\epsilon_{1,2} = \frac{\epsilon_x + \epsilon_y}{2} \pm \sqrt{\left(\frac{\epsilon_x - \epsilon_y}{2}\right)^2 + \left(\frac{\gamma_{xy}}{2}\right)^2}$$

Engineering shear strain is $\gamma_{xy} = 2\epsilon_{xy}$. For zero-temperature-change plane strain, $\sigma_z = \nu(\sigma_x + \sigma_y)$. For thermally loaded plane strain, $\sigma_z = \nu(\sigma_x + \sigma_y) - E\alpha\Delta T$. Do not set both axial strain and axial stress to zero indiscriminately.

Worked check

Gauge readings are 500, 300 and −100 microstrain at 0°,45°,90°. Then $\gamma_{xy} = 200$ microstrain. For $E = 200000$ MPa, $\nu = 0.30$, plane stress gives $\sigma_x = 103.297$ MPa, $\sigma_y = 10.989$ MPa and $\tau_{xy} = 15.385$ MPa.

When to use this model

For a general 3D stress tensor, principal stresses are the eigenvalues of its symmetric matrix. Check tensor shear symmetry and order all three principal values before evaluating a failure criterion.

Open the full chapter and practice question

Failure criteria

Symbols on this sheet

sigma1, sigma2, sigma3: principal stresses at one point; Sy: yield strength; sigma eq: chosen equivalent stress; ny: nominal yield factor. Criteria here concern ductile static yielding.

$$\sigma_{\mathrm{VM}} = \sqrt{\frac{(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2}{2}}$$

$$\sigma_{\mathrm{VM,plane}} = \sqrt{\sigma_x^2 - \sigma_x \sigma_y + \sigma_y^2 + 3\tau_{xy}^2}$$

$$\sigma_{\mathrm{Tr}} = \max |\sigma_i - \sigma_j|$$

$$n_y = \frac{S_y}{\sigma_{\mathrm{eq}}}$$

In pure shear, von Mises gives first yield at $\tau_y = S_y/\sqrt{3}$; Tresca gives $\tau_y = S_y/2$. Apply signed stresses at one point before forming the equivalent stress.

Worked check

At a shaft surface point, $\sigma_x = 100$ MPa and $\tau_{xy} = 50$ MPa, with other plane-stress components zero. Von Mises stress is 132.288 MPa. The principal stresses are 120.711, −20.711 and 0 MPa; Tresca equivalent stress is 141.421 MPa. If $S_y = 250$ MPa, yield factors are 1.890 and 1.768 respectively.

When to use this model

Static yielding is only one failure mode. Fatigue, buckling, fracture and deflection may govern even with a high calculated yield factor. A stress concentration factor depends on geometry and the definition of nominal stress.

Open the full chapter and practice question

Energy methods and impact

Symbols on this sheet

U: elastic strain energy; ks: shear correction factor; m: bending moment due to a unit virtual force. Impact W is weight (force), h is drop height and k is spring stiffness.

$$U = \int \left(\frac{N^2}{2EA} + \frac{M^2}{2EI} + \frac{T^2}{2GJ_t} + \frac{V^2}{2k_sGA} \right) dx$$

$$\delta_i = \frac{\partial U}{\partial P_i}$$

$$\theta_i = \frac{\partial U}{\partial M_i}$$

Bending unit-load method: $\delta = \int Mm/(EI)\,dx$. Ideal falling weight:

$$W(h+\delta) = \tfrac{1}{2}k\delta^2 \qquad \delta = \frac{W}{k}\left(1 + \sqrt{1 + \frac{2kh}{W}}\right)$$

For a linear spring, $U = k\delta^2/2 = P\delta/2$. An Euler-Bernoulli bending calculation omits the shear-energy term deliberately; a shear-flexible model must use a section-appropriate k_s. A common elementary rectangular-section approximation is $k_s = 5/6$.

Worked check

A uniform axial bar has $P = 10000$ N, $L = 1000$ mm, $E = 200000$ MPa, $A = 100$ mm².
$U = P^2L/(2EA) = 2500$ N·mm = 2.5 J. Differentiating gives $\delta = PL/(EA) = 0.5$ mm.

When to use this model

Impact results depend strongly on contact, mass distribution, damping and plasticity. Castigliano's simple strain-energy form here assumes linear elasticity; do not treat it as universally valid for nonlinear materials.

Open the full chapter and practice question

Indeterminate structures

Symbols on this sheet

Rj: redundant reactions; fij: displacement at coordinate i per unit force at coordinate j; Delta i: specified support displacement. Signs follow the chosen positive coordinates.

$$\delta_0 + fR = \Delta_{\text{support}} \qquad \sum_j f_{ij} R_j = \Delta_i - \delta_i^{(0)}$$

$$f_{ij} = \int \frac{m_i m_j}{EI}\, dx$$

Propped cantilever under full UDL: $R_{\text{prop}} = 3wL/8$ and fixed-end hogging moment magnitude $wL^2/8$.

Worked check

A propped cantilever of length L carries full-span downward UDL w. Remove the prop. Cantilever tip deflection magnitude from w is $wL^4/(8EI)$; an upward prop force R_B produces opposite deflection $R_B L^3/(3EI)$. Zero final tip displacement gives $R_B = 3wL/8$. Vertical equilibrium gives $R_A = 5wL/8$; the fixed-end internal bending moment is hogging with magnitude $wL^2/8$.

For $w = 2$ kN/m and $L = 4$ m, $R_B = 3$ kN, $R_A = 5$ kN and the hogging end-moment magnitude is 4 kN·m.

When to use this model

Slope-deflection, moment distribution and stiffness methods are alternative ways of enforcing equilibrium and compatible rotations/displacements. Before using any method, count independent redundants and confirm the released structure remains stable. Temperature and support settlement can cause reactions without externally applied force.

Open the full chapter and practice question

Buckling

Symbols on this sheet

Pcr: ideal critical compression; Imin: weakest relevant second moment; K: effective-length factor; lambda: slenderness; rg: radius of gyration; e: eccentricity; c: extreme-fiber distance.

$$P_{\mathrm{cr}} = \frac{\pi^2 E I_{\min}}{(KL)^2} \qquad \lambda = \frac{KL}{r_g}$$

$$\sigma_{\mathrm{Euler}} = \frac{\pi^2 E}{\lambda^2}$$

$$\sigma_{\mathrm{Johnson}} \approx S_y \left(1 - \frac{S_y \lambda^2}{4\pi^2 E}\right) \qquad \lambda_{\mathrm{transition}} = \sqrt{\frac{2\pi^2 E}{S_y}}$$

Ideal pin-ended eccentric column:

$$\sigma_{\max} = \frac{P}{A}\left[1 + \frac{ec}{r_g^2}\sec\left(\frac{L}{2r_g}\sqrt{\frac{P}{EA}}\right)\right]$$

These ideal models are not design-code column curves.

Use Johnson only on its intended branch $0 \le \lambda \le \lambda_{\mathrm{transition}}$ and Euler only in the elastic proportional range. Ideal end factors: pin-pin $K = 1$, fixed-free $K = 2$, fixed-fixed $K = 0.5$, fixed-pin $K \approx 0.699$. In the secant equation, $P > 0$ is compression, $e, c \ge 0$, equal end eccentricities are assumed and $P < P_{\mathrm{cr}}$.

Worked check

A pinned column has $E = 200000$ MPa, $I_{\min} = 10^6$ mm⁴ and $L = 2000$ mm. Euler gives $P_{\mathrm{cr}} = 493480$ N. A fixed-free column with the same data gives one-quarter: 123370 N.

When to use this model

Check elastic applicability, initial crookedness, eccentricity, residual stress, local buckling and lateral restraint. A nominal compressive stress below yield alone does not establish stability.

Open the full chapter and practice question

Unsymmetric bending

Symbols on this sheet

Iyz: centroidal product of area; u,v: principal-axis coordinates; Iu,Iv: corresponding second moments. Mu = integral(v sigma x dA); Mv = -integral(u sigma x dA).

$$I_{1,2} = \frac{I_y + I_z}{2} \pm \sqrt{\left(\frac{I_y - I_z}{2}\right)^2 + I_{yz}^2}$$

$$\sigma_x = \frac{N}{A} + \frac{M_u v}{I_u} - \frac{M_v u}{I_v}$$

Principal-axis moment definitions are in Rectangle middle-third condition: $|e| \le h/6$ for single-axis full-compression stress.

Worked check

A 100 × 200 mm rectangular section carries 100 kN compression at $e = 20$ mm along the depth. Direct compressive stress is 5 MPa. Bending stress magnitude is $Pe/Z = 3$ MPa. Edge stresses are −8 and −2 MPa, both compressive. Since $20 < 200/6$, the result agrees with the middle-third rule.

When to use this model

For thin-walled open sections, integrate shear flows and balance their moment to locate the shear center. A doubly symmetric section has its shear center at the symmetry intersection; a channel generally does not have it at its centroid.

Open the full chapter and practice question

Curved beams

Symbols on this sheet

r: radius from initial center of curvature; rn: neutral radius; rc: centroid radius; e = rc - rn; A: area. Positive M causes inner-fiber tension in the declared convention.

$$r_n = \frac{A}{\int_A dA/r} \qquad r_c = \frac{\int_A r\,dA}{A}$$

$$e = r_c - r_n \qquad \sigma_\theta = \frac{M}{Ae}\left(\frac{r_n}{r} - 1\right)$$

Positive M produces inner-fiber tension under the defined pure-bending convention. For constant width, $r_n = (r_o - r_i)/\ln(r_o/r_i)$.

Worked check

Let $b = 20$ mm, $r_i = 40$ mm, $r_o = 80$ mm and $M = 100000$ N·mm, with no direct force. Then $A = 800$ mm², $r_c = 60$ mm, $r_n = 57.708$ mm and $e = 2.292$ mm. Inner stress is about +24.141 MPa and outer stress −15.196 MPa.

When to use this model

This is curved-beam theory, not a complete hook design. Loads near contact zones, transverse shear, ring constraints and three-dimensional effects require further analysis. For a hook, calculate both direct force and bending at the cut. Avoid subtractive roundoff when r_c and r_n nearly coincide in a very shallow curved section.

Open the full chapter and practice question

Thick cylinders

Symbols on this sheet

a,b: inner/outer radii; pi,po: positive pressure magnitudes; C,D: Lame constants; r: wall radius; ur(r): radial displacement. Pressure gives compressive radial stress.

$$C = \frac{p_i a^2 - p_o b^2}{b^2 - a^2} \qquad D = \frac{a^2 b^2 (p_i - p_o)}{b^2 - a^2}$$

$$\sigma_r = C - \frac{D}{r^2} \qquad \sigma_\theta = C + \frac{D}{r^2}$$

For pressure-loaded closed ends, $\sigma_z = C$. Plane-stress displacement: $u_r(r) = [(1-\nu)Cr + (1+\nu)D/r]/E$.

With a uniform specified axial stress:

$$u_r(r) = \frac{[(1-\nu)C - \nu\sigma_z]r + (1+\nu)D/r}{E}$$

Thus closed ends carrying pressure thrust give

$$u_r(r) = \frac{(1-2\nu)Cr + (1+\nu)D/r}{E}$$

For zero axial strain with no temperature change, $\sigma_z = 2\nu C$, a different condition. Always check $\sigma_r(a) = -p_i$ and $\sigma_r(b) = -p_o$.

Worked check

$a = 50$ mm, $b = 100$ mm, $p_i = 10$ MPa, $p_o = 0$. Then $C = 3.333$ MPa, $D = 33333.333$ MPa·mm². At the bore, $\sigma_r = -10$ MPa and $\sigma_\theta = 16.667$ MPa. At the outside, they are 0 and 6.667 MPa. Closed-end axial stress is 3.333 MPa.

When to use this model

A shrink fit creates contact pressure from radial compatibility: outer-member bore expansion minus inner-member surface expansion equals radial interference. Distinguish radial from diametral interference. Include both parts' elastic response and superpose later pressure loads only while contact and elasticity assumptions remain valid. External-pressure instability needs a separate check.

Open the full chapter and practice question

Noncircular torsion

Symbols on this sheet

Jt: torsion constant, not generally the polar moment; b,t: strip width/thickness; Am: median-line enclosed area; q: circulating shear flow; s: wall arc length.

Thin strip: $J_t \approx bt^3/3$, $\tau_{\max} \approx 3T/(bt^2)$. Open strips: $J_t \approx \sum b_i t_i^3/3$. Single closed cell:

$$q = \frac{T}{2A_m} \qquad \tau = \frac{q}{t}$$

$$\phi' = \frac{T}{4A_m^2}\oint \frac{ds}{Gt} \qquad J_t = \frac{4A_m^2}{\oint ds/t} \quad \text{for constant } G$$

Worked check

A rectangular single-cell tube has median dimensions 100 × 50 mm and uniform thickness 2 mm. $A_m = 5000$ mm² and median perimeter 300 mm. At $T = 10^6$ N·mm, $q = 100$ N/mm and $\tau = 50$ MPa. $J_t = 666666.7$ mm⁴. For $G = 80000$ MPa and $L = 1000$ mm, twist is 0.01875 rad.

When to use this model

Multiple closed cells need torque equilibrium and equal compatible twist, with differences of cell flows on common walls. Restrained warping adds longitudinal normal stresses and cannot be captured by free-warping Saint-Venant torsion alone.

Open the full chapter and practice question

Fatigue, fracture and creep

Symbols on this sheet

Sa (written sigma a): amplitude; sigma m: mean; Se: appropriate component fatigue strength; Sut: tensile ultimate strength; Y: crack geometry factor; a: specified crack size; T: kelvin in creep.

$$\sigma_a = \frac{\sigma_{\max} - \sigma_{\min}}{2} \qquad \sigma_m = \frac{\sigma_{\max} + \sigma_{\min}}{2}$$

$$\frac{\sigma_a}{S_e} + \frac{\sigma_m}{S_{ut}} \le \frac{1}{n}$$

$$\mathcal{D} = \sum \frac{n_i}{N_i} \qquad K_f = 1 + q_n(K_t - 1)$$

$$K_I = Y\sigma\sqrt{\pi a} \qquad a_c = \frac{1}{\pi}\left(\frac{K_{IC}}{Y\sigma}\right)^2$$

$$\frac{da}{dN} = C_P(\Delta K)^m \qquad \dot{\epsilon} = A_c\sigma^n \exp[-Q_c/(RT)]$$

These are specific empirical or idealized models, not universal material properties.

Stress ratio and a fitted stress-life curve:

$$R_\sigma = \frac{\sigma_{\min}}{\sigma_{\max}} \qquad \sigma_a = C_f N_f^{b_f}$$

Do not evaluate the ratio when $\sigma_{\max} = 0$. The displayed Goodman screening equation uses tensile mean stress, appropriate component fatigue data and the stated safety-factor convention. For constant crack-geometry factor Y, the explicit a_c expression applies; otherwise solve $Y(a_c)\sigma\sqrt{\pi a_c} = K_{IC}$. Paris and creep constants are unit-, temperature- and material-dependent.

Worked check

With $\sigma_{\max} = 120$ MPa and $\sigma_{\min} = 20$ MPa, amplitude is 50 MPa and mean is 70 MPa. For assumed component data $S_e = 200$ MPa and $S_{ut} = 500$ MPa, Goodman utilization is $0.25 + 0.14 = 0.39$, giving $n = 2.564$ in that model.

When to use this model

Do not confuse stress intensity K_I with concentration K_t. Static formulas cannot establish fatigue life without load history and suitable material/environment data.

Open the full chapter and practice question

Plastic bending

Symbols on this sheet

Ze,Zp: elastic and plastic section moduli; yp: plastic neutral axis; My,Mp: first-yield/fully-plastic moments; Sy: ideal symmetric yield strength. Pure bending is assumed.

$$M_y = S_y Z_e \qquad M_p = S_y Z_p$$

$$Z_p = \int_A |y - y_p| dA$$

Rectangle: $Z_e = bh^2/6$, $Z_p = bh^2/4$, shape factor 1.5. Simply supported central-force ideal collapse: $P_c = 4M_p/L$.

Plastic shape factor is Z_p/Z_e. At an ideal collapse mechanism, external work equals $\sum M_p|\Delta\theta|$ over the plastic hinges. Rotation capacity and absence of premature instability are necessary assumptions.

Worked check

A 40 × 80 mm rectangle with $S_y = 250$ MPa has $Z_e = 42666.7$ mm³ and $Z_p = 64000$ mm³. First-yield moment is 10.667 kN·m; fully plastic moment is 16 kN·m.

For a simply supported span with a central point force, the ideal bending collapse load is $P_c = 4M_p/L$ if shear and instability do not intervene.

When to use this model

Unloading after partial yielding can leave residual stresses and permanent curvature. Local buckling, lateral-torsional buckling, brittle fracture or inadequate rotation capacity can prevent development of M_p. A nominal plastic hinge is a model of a yielded region, not a physical pin installed in the beam.

Open the full chapter and practice question

Composites

Symbols on this sheet

Ei,Ai,Ici: layer modulus, area and centroidal second moment; yE: elastic neutral axis; Vf,Vm: fiber/matrix volume fractions; E1,E2,G12,nu12,nu21: orthotropic lamina properties.

$$y_E = \frac{\sum E_i A_i y_i}{\sum E_i A_i}$$
$$(EI)_{\text{eff}} = \sum E_i [I_{ci} + A_i (y_i - y_E)^2]$$
$$\sigma_i = -E_i \frac{M}{(EI)_{\text{eff}}} (y - y_E)$$

$$E_L \approx V_f E_f + V_m E_m \qquad E_T \approx \left(\frac{V_f}{E_f} + \frac{V_m}{E_m} \right)^{-1}$$
$$\frac{\nu_{12}}{E_1} = \frac{\nu_{21}}{E_2}$$

For the ideal orthotropic plane-stress lamina:

$$\epsilon_1 = \frac{\sigma_1}{E_1} - \frac{\nu_{21}\sigma_2}{E_2} \qquad \epsilon_2 = -\frac{\nu_{12}\sigma_1}{E_1} + \frac{\sigma_2}{E_2}$$
$$\gamma_{12} = \frac{\tau_{12}}{G_{12}}$$

For bonded beams, curvature is $\kappa = M/(EI)_{\text{eff}}$. Perfect bond and plane-section kinematics are required.

Worked check

For $V_f = 0.6$, $E_f = 200$ GPa and $E_m = 3$ GPa, the longitudinal estimate is 121.2 GPa. The reciprocal transverse estimate is 7.335 GPa. Direction therefore matters enormously even with identical constituents.

When to use this model

Layered laminates may couple extension, bending and twist. Slip, delamination, matrix cracking and directional strengths require additional models. Perfect-bond transformed-section calculations cannot detect interface failure by themselves.

Open the full chapter and practice question

Elasticity and bar finite elements

Symbols on this sheet

u: displacement vector; b: body force per volume; C: elasticity tensor; I: identity tensor; ke,K: element/global stiffness matrices; F: load vector. The bar element is a 1D idealization.

$$\nabla \cdot \boldsymbol{\sigma} + \mathbf{b} = 0 \qquad \boldsymbol{\epsilon} = \tfrac{1}{2}[\nabla \mathbf{u} + (\nabla \mathbf{u})^T]$$
$$\boldsymbol{\sigma} = \mathbf{C} : \boldsymbol{\epsilon}$$

$$\mathbf{k}_e = \frac{EA}{L}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \qquad \mathbf{K}\mathbf{u} = \mathbf{F}$$
$$\epsilon_e = \frac{u_2 - u_1}{L}$$

The stress-strain equation above excludes initial/thermal strain. With isotropic thermal expansion:

$$\boldsymbol{\sigma} = \mathbf{C} : (\boldsymbol{\epsilon} - \alpha \Delta T \mathbf{I})$$

Apply sufficient independent displacement constraints before solving the assembled stiffness system.

Worked check

One bar has $E = 200000$ MPa, $A = 100$ mm² and $L = 1000$ mm. Fix node 1 and apply 10000 N at node 2. The reduced stiffness is 20000 N/mm, so $u_2 = 0.5$ mm. Strain is 0.0005, stress is 100 MPa and the support reaction balances −10000 N.

When to use this model

Check units, reaction equilibrium, deformed shape, material assignment and energy. Refine the mesh and compare a chosen output rather than only looking at stress colors. Point loads, sharp re-entrant corners and ideal restraints can produce nonconvergent peak stresses; choose physically meaningful evaluation locations.

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Sources and verification

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